

Research paper

Location optimization of EV charging stations: A custom K-means cluster algorithm approach

Muhammad Rabiul Abdullahi^{a,b}, Qing-Chang Lu^a, Adil Hussain^{a,*}, Sajib Tripura^{a,c}, Peng-Cheng Xu^a, ShiXin Wang^a

^a Department of Traffic Information and Control Engineering, School of Electronics and Control Engineering, Chang'an University, Xi'an 710064, China

^b Department of Electrical Electronic Engineering, Collage of Engineering, Kaduna Polytechnic, Kaduna 800262, Nigeria

^c Department of Computer Science and Engineering, Rangamati Science and Technology University, 4500, Bangladesh

ARTICLE INFO

Keywords:

EV
Charging station
Custom K Means
Penalty term
Optimization
Energy consumption

ABSTRACT

The strategic deployment of EV (EV) charging stations is crucial for promoting sustainable transportation and facilitating the widespread adoption of EVs. However, the lack of readily accessible charging station continues to be a significant barrier to the mainstream adoption of EVs. This study presents a comprehensive approach to optimizing the location of charging stations using a custom K-means clustering algorithm. The algorithm incorporates various factors including charging demand, energy consumption, population density, and existing stations, to identify optimal locations for the charging stations. The proposed methodology aims to minimize the distance between charging stations and areas with high EV demand while considering energy efficiency and avoiding redundancy near low-weighted point locations. The algorithm iteratively assigns data points to clusters and updates the centroids, converging to an optimal solution that balances coverage, accessibility, and sustainability. Folium map in Python and Python code was used for case study analysis. The innovation of the finding shows the effectiveness of the custom K-means clustering algorithm in optimizing charging station placement, by introducing a penalty term to avoid repositioning charging stations close to a low-weighted (charging demand) point charging station, also ensuring convenient access for EV owners, and promoting efficient energy utilization. The study emphasizes on strategic planning and data-driven approaches in developing a comprehensive and sustainable EV charging station network to address the challenge of limited charging station and support the growth of electric mobility.

1. Introduction

The Chinese government's Guidelines for the Development of EV Charging Station (2015–2022) include a thorough strategy to establish 5 million charging piles and over 12,000 charging stations by 2022 to meet the charging requirements of the nation's 5 million EVs (Yang et al., 2020). Nonetheless, certain disadvantages may hinder EV adoption, including prolonged recharging durations, elevated costs, and, most critically, the insufficient number of charging stations. A survey conducted in the USA revealed that approximately 83 % of respondents identified charging stations as their primary concern (Sadeghi Ahangar et al., 2022). According to current EV development policies promulgated by governments globally, there will be 14 million slow charging stations and 2.3 million fast public charging stations by 2030 (Afshar et al., 2021). A Deloitte survey indicates that 78 % of prospective EV

purchasers in most advanced nations express concern regarding the accessibility of charging stations.

Concerns regarding climate change and air pollution, coupled with advancements in battery technology and governmental incentives, have contributed to the consistent increase in the adoption of EVs in recent years. The absence of easily available charging station hinders the widespread adoption of EVs (Carra et al., 2022). The location and capacity of charging stations significantly influence the accessibility and ease of charging for EV owners, hence affecting the adoption of EVs. The storage and delivery of supplies are the primary distinctions between electric transport and current carbon-based systems. Power grids function as real-time systems for supply and demand, characterised by limited storage capacity and stringent supply-side control (I. Safak Bayram, S. B. 2020).

The purported chargers Level 1, 2, and 3 exhibit power levels ranging

* Corresponding author.

E-mail address: 2022032907@chd.edu.cn (A. Hussain).

<https://doi.org/10.1016/j.egy.2024.09.075>

Received 12 July 2024; Received in revised form 16 September 2024; Accepted 30 September 2024

Available online 15 November 2024

2352-4847/© 2024 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

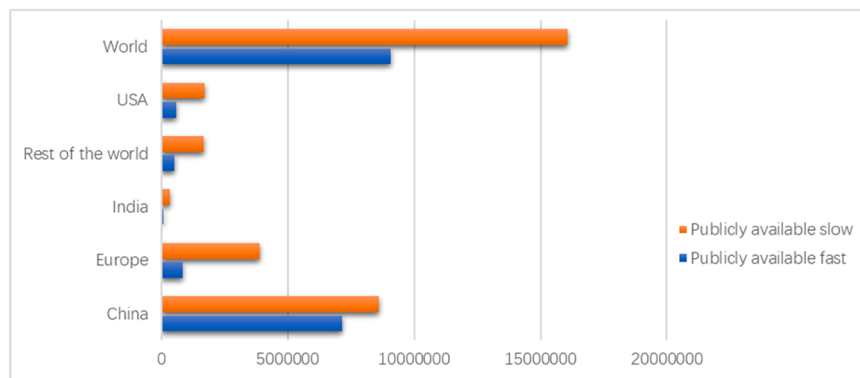


Fig. 1. World EVCS adoption.

from 1.4 kW, 7.7 kW, and 13.3 kW to 1.9 kW, 25.6 kW, and 96 kW, respectively. Moreover, the standard charging durations for Level 1, Level 2, and Level 3 chargers are 11.5 hours, 2 hours, and 0.5 hours, respectively. Consequently, Level 1 chargers are more appropriate for residential charging, whereas Level 2 and 3 chargers are more suitable for public or private facilities (Bouguerra and Bhar Layeb, 2019). The three fundamental methods of EV charging are conductive charging, inductive charging, and battery replacement (Erbaş et al., 2018). The conductive charging method involves utilising a cable to link the battery to the electrical source. Despite the absence of physical contact between the EV and the charging station, the inductive method functions through electromagnetic transmission. The third option entails replacing the depleted batteries in a station with new ones (Erbaş et al., 2018).

The feasibility of long-distance travel with EVs is influenced by the capacity of charging stations. The IONITY network of high-power charging stations has been built to offer reliable charging options for long-distance travel, with plans for extensive station construction in the coming years (Afshar et al., 2021; Liu et al., 2022). Fig. 1 illustrates the global availability of EV charging stations, with China exhibiting the biggest number of both slow and fast charging stations, followed by Europe, the USA, the rest of the globe, and finally India.

Comprehending the elements that influence the convenience and efficiency of charging stations, such as charging demand, proximity and providers of EV charging stations, informed decisions can be made regarding the optimal placement of new stations and the enhancement of existing ones. Increased availability and efficiency of EV charging stations may reduce greenhouse gas emissions and improve air quality. This research may facilitate the attainment of those objectives.

The main contributions of this research work are as follows:

- This study proposed the custom K-means cluster algorithm for optimization of both the existing stations and placement of repositioned stations.
- Optimizing the existing charging station using custom K-Means algorithm based on the charging demand and proximity also introducing a penalty term for the recommended charging station and also show the impact of penalty term on the existing charging station and show the optimal location for the recommended charging station with and without the penalty term.
- The proposed algorithm has been compared with the general k-means algorithm.

This paper is organized as follows: Section II contains the Literature, Section III contains the Method Design includes the data analysis and implementation, Section IV is the results and analysis. Section V is the conclusion.

2. Literature review

In The establishment of a comprehensive network of charging stations is essential for the extensive adoption of EVs (EVs). Governments and private enterprises globally have started investing in charging stations to address this need (Savari et al., 2023). Numerous EV owners opt to install charging stations at their residences, in addition to public charging stations. This provides a pragmatic and reliable approach for charging automobiles overnight and may help alleviate the strain on public charging stations during peak demand periods (Lu et al., 2021; Mastoi et al., 2022). Most industrialised countries, such as China, Bulgaria, and Brazil, mostly utilise grid-tied systems, whereas India, Romania, and Vietnam employ off-grid systems, and Qatar operates on a stand-alone mode (Oladigbolu et al., 2022).

As battery technology progresses, the proliferation of charging stations increases, and consumer demand for environmentally friendly transportation options escalates, the trend towards EVs is expected to persist (Mishra et al., 2021). A study by (I. Safak Bayram, S. B. 2020) indicates that standard operations research approaches effectively solve coverage difficulties, while queueing models are applicable for capacity planning and determining whether charging facilities should be located as clusters or interconnected regions.

Limited accessibility may have necessitated detours for cars, increasing their journey distance to the charging station and thereby worsening the user experience. Determine optimal locations for the installation of additional charging stations based on the daily travel and dwelling habits of a taxi fleet (Yang et al., 2017).

Given the capacity of charging stations and its impact on EV adoption, prior research and algorithms suggest that potential EV buyers may be deterred by insufficient charging station. High-capacity charging stations alleviate range anxiety and enhance consumer interest in EVs (Yang et al., 2017). An EV battery can attain 80 % capacity using a fast-charging station providing 150–350 kW in around 15–30 minutes, significantly minimising the charging duration (Afshar et al., 2021; Bouguerra and Bhar Layeb, 2019).

2.1. Optimisation models and algorithms

In (Zhou et al., 2022) elucidate the ideal distribution of charging stations by a genetic algorithm technique, providing a theoretical foundation for simulating their deployment. However, neglect some aspects affecting charging station placement, the assessment of user behaviour unpredictability, and the incorporation of strategic behaviour and load variability in the planning of EV charging stations. A study by (Qiao et al., 2018) indicates that the likelihood of utilising public charging stations is 39.6 %, suggesting that battery EV (BEV) operators favour home charging. The study also employed an optimisation algorithm utilising a Voronoi diagram to identify potential locations for new charging stations; however, it neglects to address the environmental

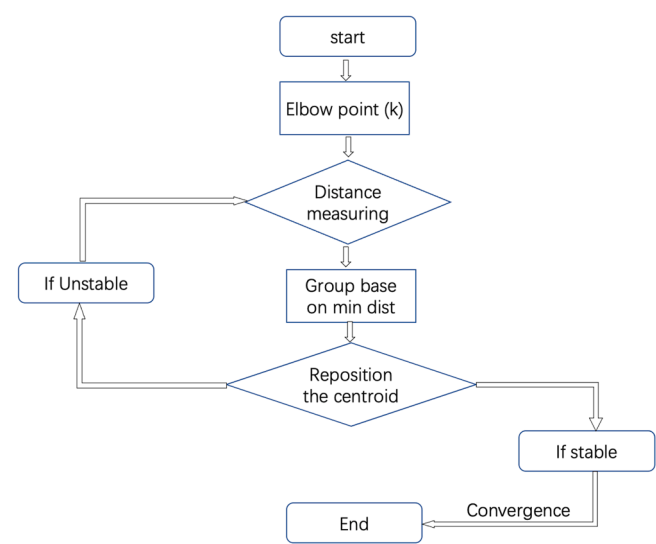


Fig. 2. K-means cluster algorithm flow chart.

impact or sustainability considerations associated with the deployment of these stations. A subsequent study by (He et al., 2015), utilising the Sioux Falls network, determined that it incorporates various initial states of charge and drivers’ risk-taking attitudes. Additionally, (Oladigbolu et al., 2022) discuss that Nigeria’s optimal configuration for

renewable energy-based EV charging stations is a photovoltaic/wind turbine/battery system in Sokoto, which minimises carbon emissions and promotes EV development. Both (He et al., 2015 and Oladigbolu et al., 2022) neglect to account for the effects of various charging technologies, overlook spontaneous alterations in travel and recharging decisions, and fail to investigate the integration of EV charging stations with the current power grid station, considering the associated social and environmental implications.

A study conducted by (Uslu and Kaya, 2021) aimed to reduce billing costs in a municipal electricity grid through resource sharing optimisation. The emphasis is on organising a substantial quantity of EVs at a charging station with a distribution algorithm queuing approach and mixed integer programming. These algorithms include aspects such as the available State of Charge (SOC) of the cars, the duration needed to achieve full battery capacity, and the utility cost. The charging rate is regarded as a continuous variable to attain optimal cost reduction. Key findings from the meso-level approach indicate that taxi fleet dwell patterns influence the placement of charging stations. Enhancing waiting space increases charger utilisation but necessitates additional parking policies and operational recommendations, providing a practical foundation for cities to implement EV Charging Stations (EVCS) (Torkey and Abdelgawad, 2022). Further research is required to address the complexities of EVCS planning and its interplay with various operational factors, including driving and charging behaviour, ambient temperature, and traffic conditions.

Consequently, (Torres Franco et al., 2021) demonstrate that a distinctive application of the chicken swarm algorithm indicates that

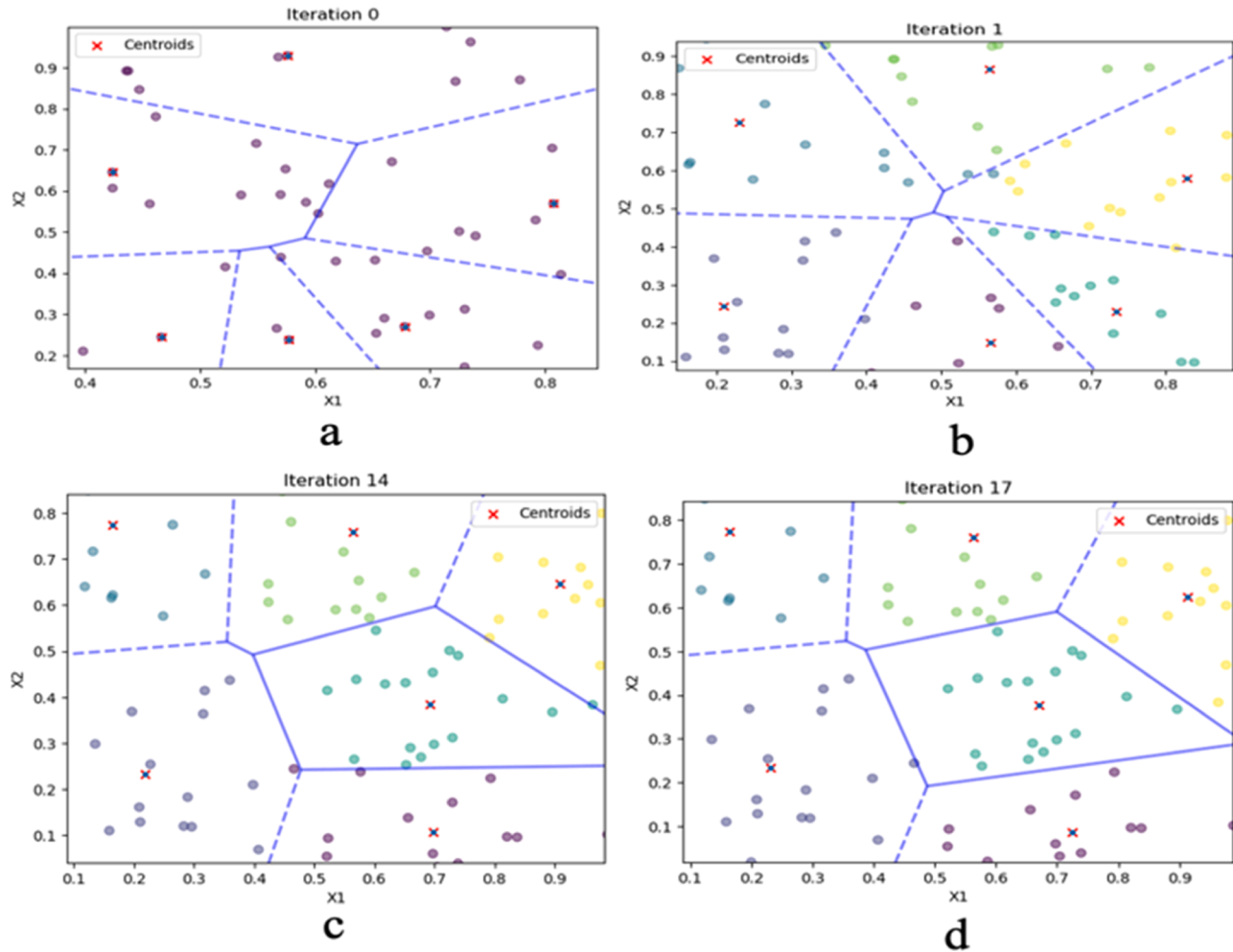


Fig. 3. Processing sequence of K-Mean algorithm.

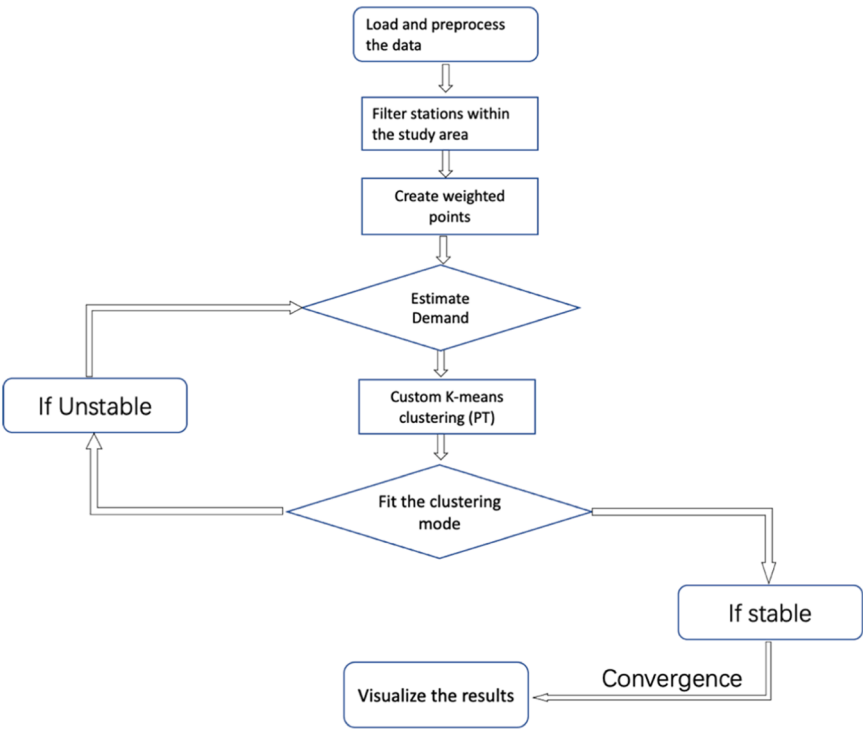


Fig. 4. Implementation flow chart using Custom K-means cluster algorithm.

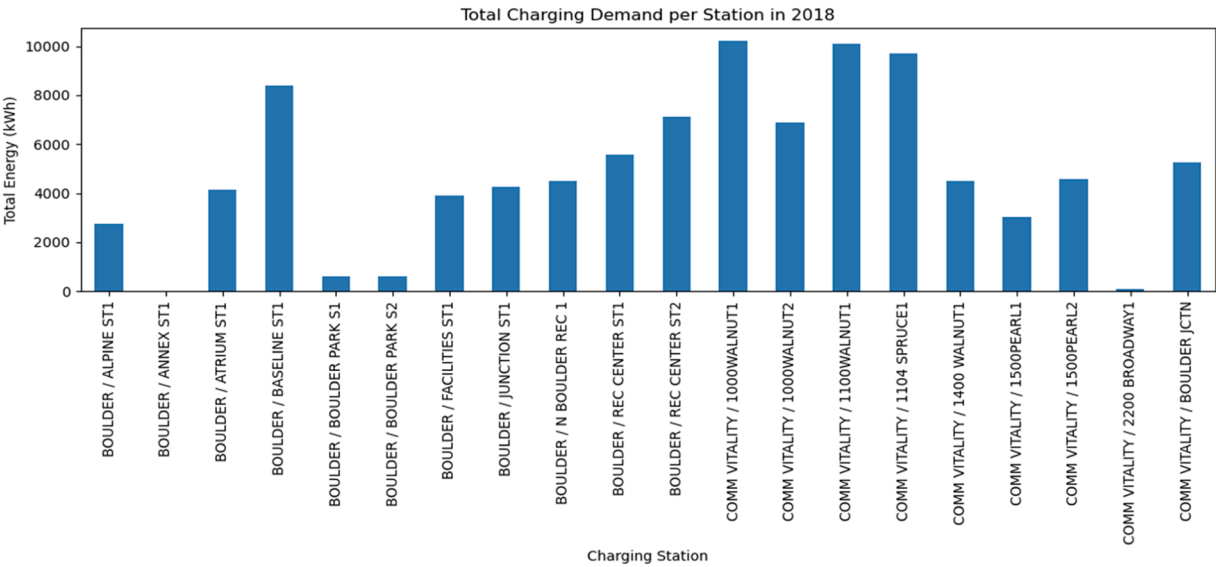


Fig. 5. Energy Consumption per Kwh for each station in 2018.

driving ranges are paramount for the efficient utilisation of electric buses. Furthermore, variables such as charging durations, trip frequency, and service rates substantially influence station capacities, yet there is limited discourse on the explicit consideration of waiting times at charging stations, a critical element for passenger satisfaction and operational efficiency.

A further proposed approach by (Shareef et al., 2016 and Yang et al., 2020) entails optimising the scheduling of EVs with stochastic arrival times. The objective is to reduce charging expenses by taking into account variables such as battery capacity, available State of Charge (SOC), charging duration, arrival and departure times, and charging rates. The cost of electric power is presumed to be uniform at all charging sites. Demand forecasts are transmitted from a central

controller to the local controller, which subsequently employs an optimisation algorithm to schedule the EVs for charging or discharging. Recent research by (Zheci Tang et al., 2013) demonstrated that charging stations situated at park-and-ride facilities, near concentrated services, and in high-density regions are more effective in meeting urban public charging demand than petrol stations. To enhance the precision of charging demand estimation, detailed data on travel patterns and real-time traffic information are essential. Additional results derived from the analysed scenarios indicated that the effects on power system parameters (Kong et al., 2019) necessitate the development of new optimisation algorithms or strategies to enhance the efficiency and precision of EVCS placement. (Chakraborty et al., 2022) present a systematic methodology for site selection, enhancing the decision-making

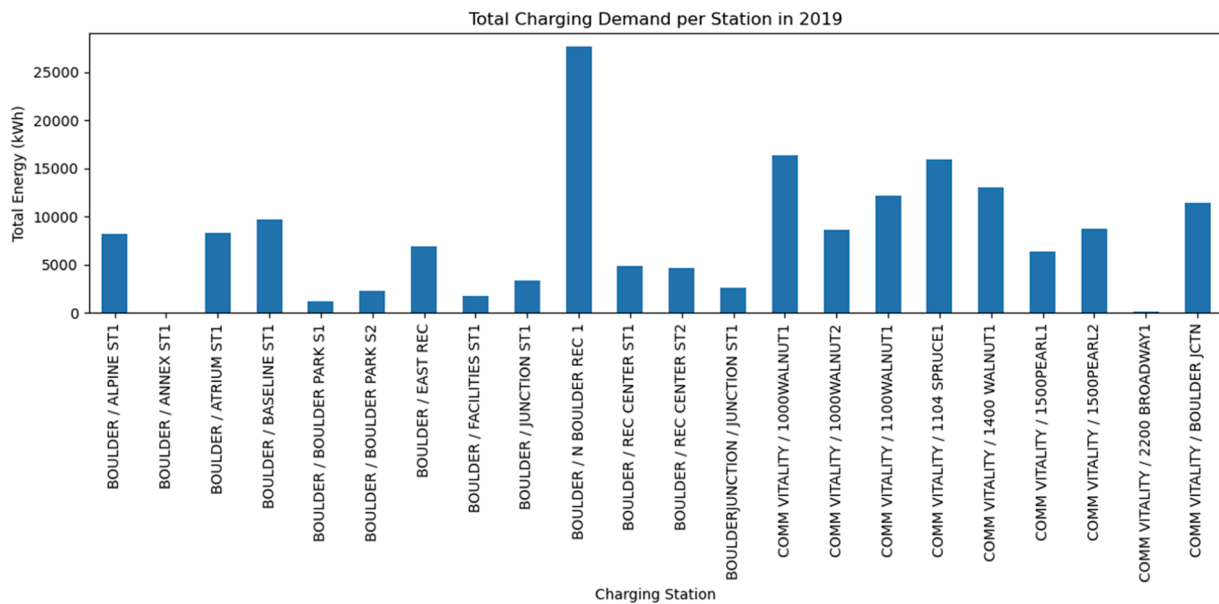


Fig. 6. Energy Consumption per Kwh for each station in 2019.

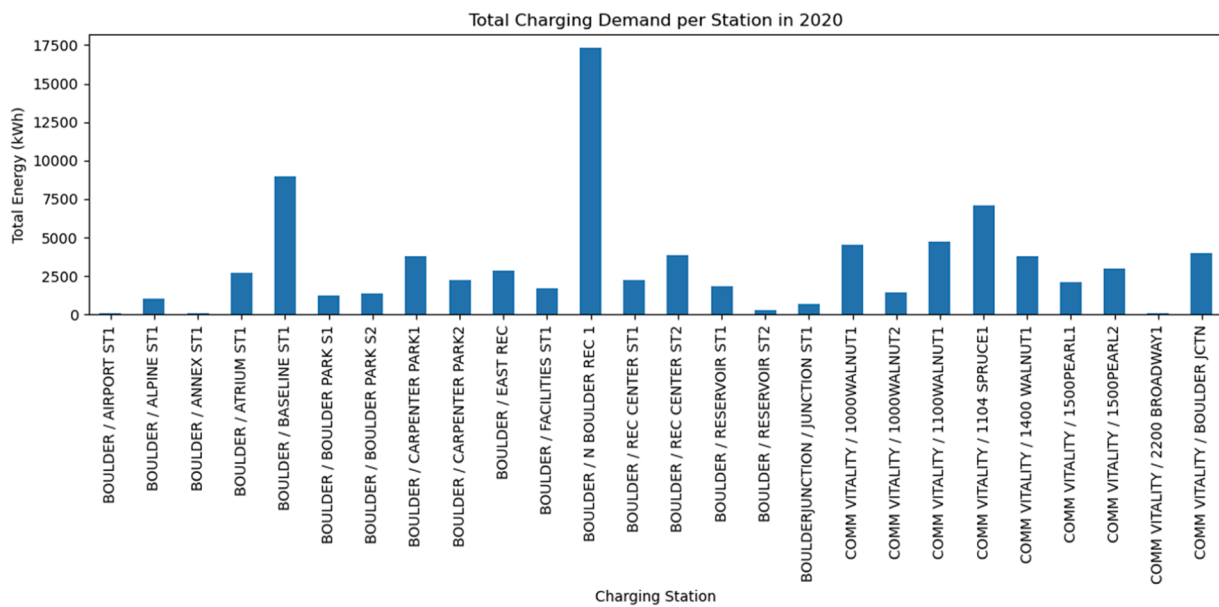


Fig. 7. Energy Consumption per Kwh for each station in 2020.

process for establishing EV charging stations. Given the expansion of EVs, a comprehensive approach is essential to support extensive networks of charging stations.

The findings of the study by (Bitencourt et al., 2021) proposed a methodology to identify optimal locations for mid-term and long-term EV penetration, taking into account both technical and mobility factors. It emphasises the necessity of addressing long-term planning through a multi-objective approach while considering uncertainties associated with EV users. The optimal location planning aims to maximise operators' economic interests, improve drivers' charging satisfaction, and ensure traffic efficiency and power grid safety; however, more comprehensive studies that integrate all relevant factors and employ dynamic data for precise location planning of fast charging stations for EVs are required (Kong et al., 2019).

A sophisticated scheduling technique proposed by (Yang et al., 2017 and Zhan et al., 2022) aims to minimise expenses for EVs by decreasing

waiting times in scenarios when charging station availability is constrained. A simulation is run to ascertain the optimal position for charging stations on a roadway for EVs. Nonetheless, it is crucial to acknowledge that it predominantly emphasises travel duration and neglects fluctuations in EV energy usage and proximity, which are affected by the velocity of the EV.

Recently, (Bui et al., 2023) conducted an in-depth study on the optimal operation of charging stations amidst rising EV penetration, employing the K-means clustering method for EVs and a mixed-integer linear programming (MILP) approach to maximise charging station profits while minimising imported electricity.③④⑤⑥⑦⑧⑨

The application of data analysis and modelling to ascertain optimal sites for charging stations and establish the requisite capacity for each site is essential. Areas with high population density necessitate larger capacity charging stations to meet demand (Carra et al., 2022; Chen et al., 2020). Regions with a scarcity of existing charging stations may

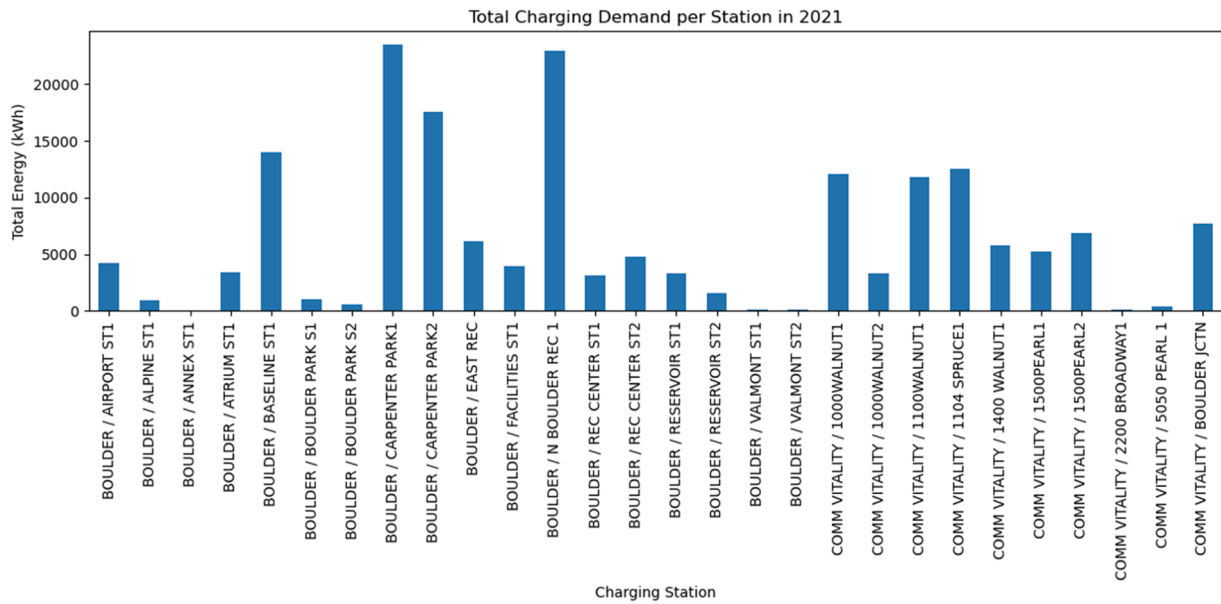


Fig. 8. Energy Consumption per Kwh for each station in 2021.

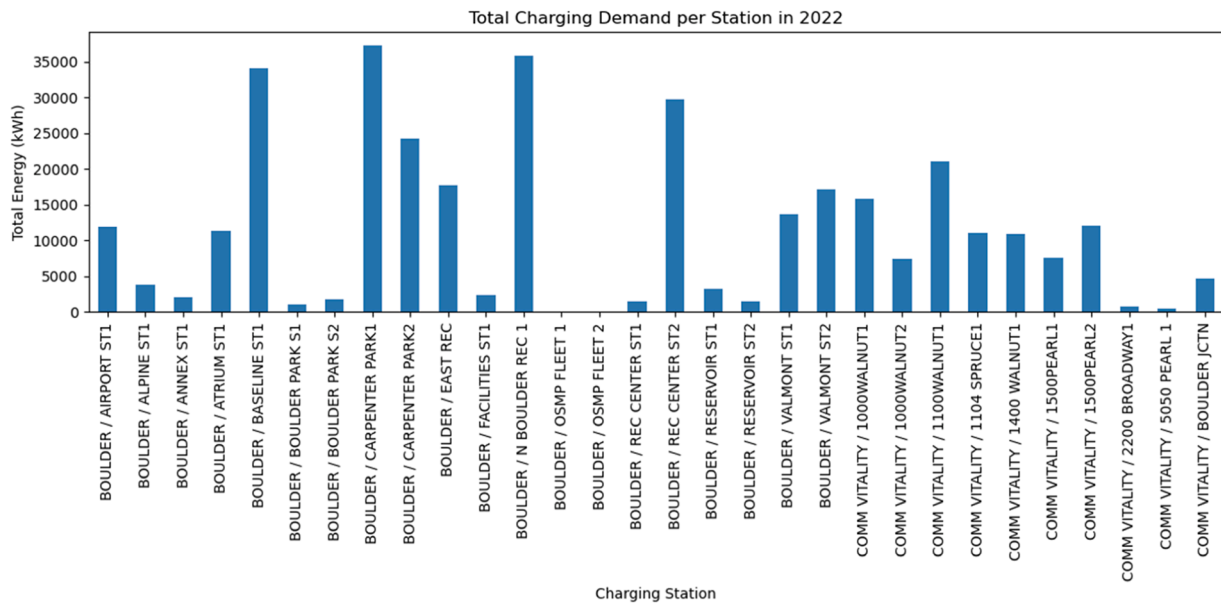


Fig. 9. Energy Consumption per Kwh for each station in 2022.

also require larger capacity stations to offset this deficiency (Guler and Yomralioglu, 2020). Furthermore, locations with significant traffic flow may demand larger capacity charging stations to accommodate peak hour demand (Afshar et al., 2021). The energy requirements of vehicles utilising charging stations must be taken into account when assessing the suitable capacity of these stations (Sadeghi Ahangar et al., 2022). Various regions may possess differing energy generation capabilities, influencing the optimal capacity of the charging stations (Guler and Yomralioglu, 2020).

Previous studies examine many features of EV charging stations and propose diverse optimisation strategies for their development and operation. It underscores the significance of charging stations in facilitating extensive EV adoption and the advantages of their installation at residential and essential locations. We identified several research needs, including the necessity to account for aspects such as locational influences, fluctuating demand, travel patterns, user behaviour

variability, strategic actions, and load uncertainty in the development and optimisation of charging stations. Moreover, it underscores the necessity of tackling challenges such as restricted accessibility, prolonged waiting periods at charging stations, and the incorporation of EV charging stations into the current power grid framework.

Furthermore, new studies indicate the necessity for detailed data to accurately estimate charging demand for the optimisation of existing charging stations. The studies offer numerous optimisation approaches and insights into the placement and operation of charging stations, highlighting the necessity for additional study to fill the highlighted gaps for more efficient and effective EV charging station placement.

Based on the aforementioned literature, this research proposes the utilisation of a folium map to delineate the study area, identify existing charging stations, illustrate the road network of these stations, and provide a visual representation of both the current and proposed locations for the optimal placement of a new charging station. Employing a

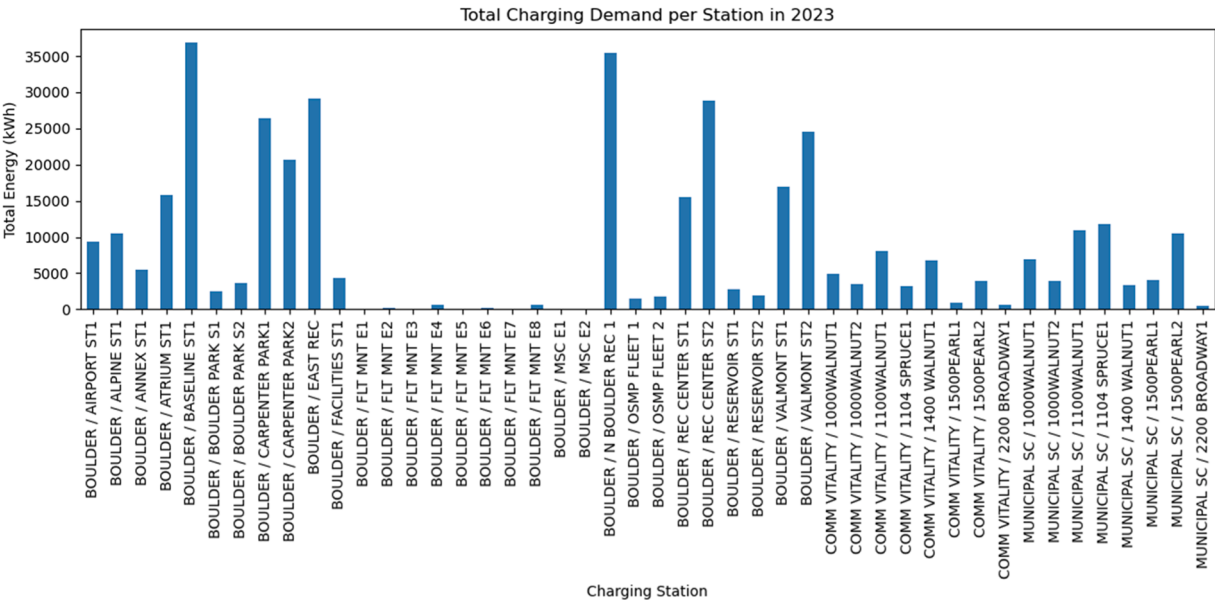


Fig. 10. Energy Consumption per Kwh for each station in 2023.

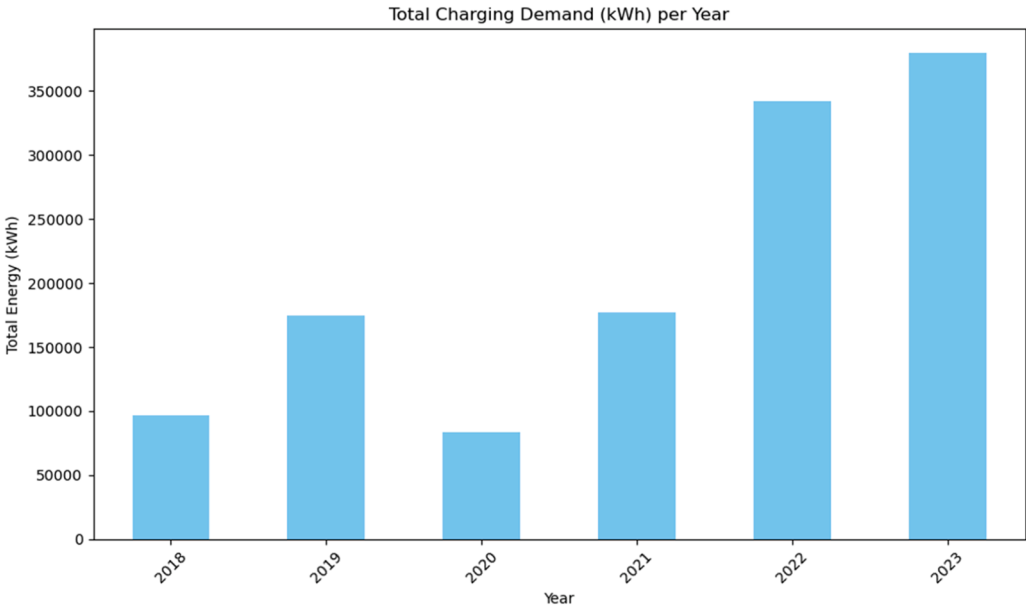


Fig. 11. Energy consumption from 2018 to 2023.

Custom K-means clustering method to optimise the current charging stations and determine the ideal placement of relocated stations, taking into account charging demand and incorporating a penalty term to prevent the relocation of stations near low-weighted existing locations.

3. Design and optimisation of models

3.1. Analysis of Geographic Information System (Folium map)

Identify the charging stations and evaluate geographical data utilising Geographic Information System (folium map) technology, including characteristics such as the road network to key destinations, employing the state algorithm to determine optimal locations for charging stations. Employ the data to ascertain ideal sites for charging stations, informed by historical data and future growth forecasts.

3.2. General concept of k-means clustering algorithm

The k-means algorithm is a widely utilised unsupervised machine learning method for categorising data points into discrete groups according to their similarity. The objective is to divide the data into k clusters, with k being a user-defined parameter.

The algorithm operation:

- Initialization: Randomly select k points from the dataset to serve as the first cluster centroids.
- Task: For every data point, compute its distance to each centroid and allocate it to the closest centroid, so establishing initial clusters.
- Revise the centroids of the clusters by computing the average of all data points allocated to each cluster.
- Iterate: Repeat steps 2 and 3 until convergence is attained. Convergence transpires when the centroids cease to exhibit substantial

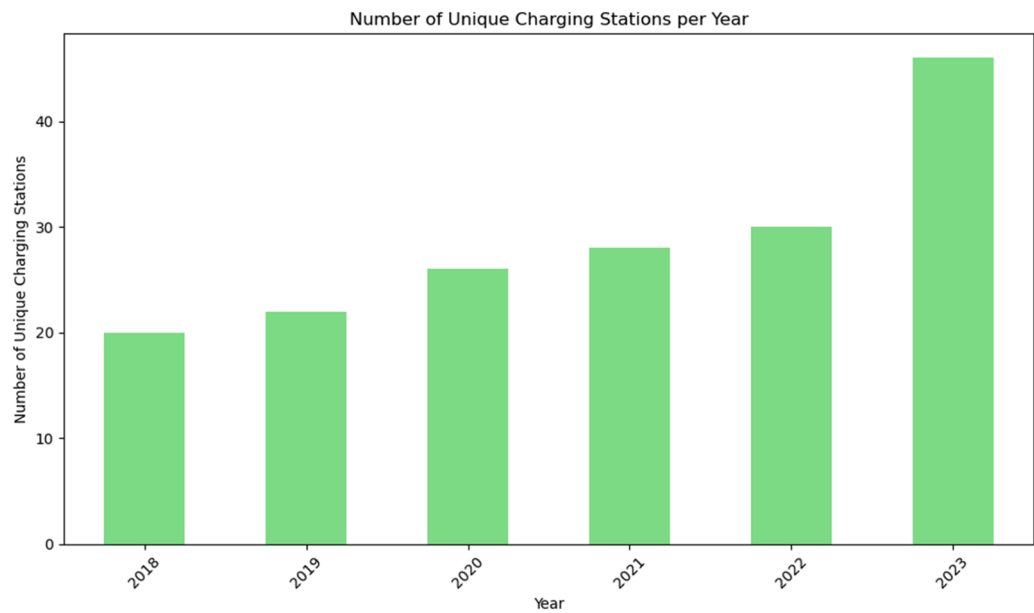


Fig. 12. Number of charging station from 2018 to 2023.

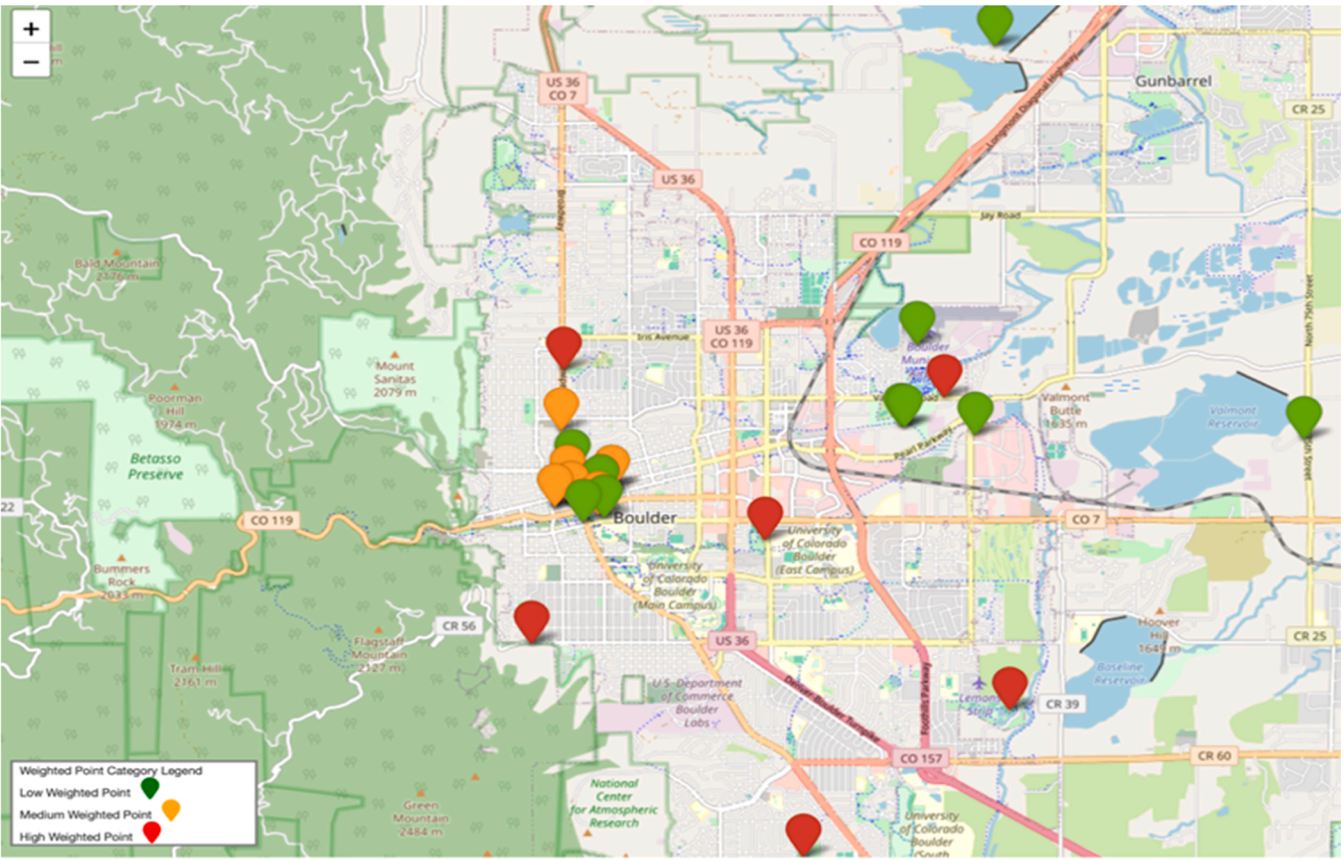


Fig. 13. locations of the existing stations.

changes or when a predetermined maximum number of repetitions is attained.

- The algorithm produces the final k clusters, assigning each data point to the cluster corresponding to the nearest centroid.

Fig. 2 illustrates the objective of minimising the within-cluster sum of squares, referred to as inertia or distortion. It is intended to be

accomplished by the iterative optimisation of cluster centroids. The algorithm converges when the centroids stabilise and the data points are allocated to their optimal clusters.

The k -means method necessitates that the user predefine the number of clusters, k . Selecting a suitable value of k is frequently subjective and can influence the quality of clustering outcomes. Techniques such as the elbow approach and silhouette analysis are available to ascertain the

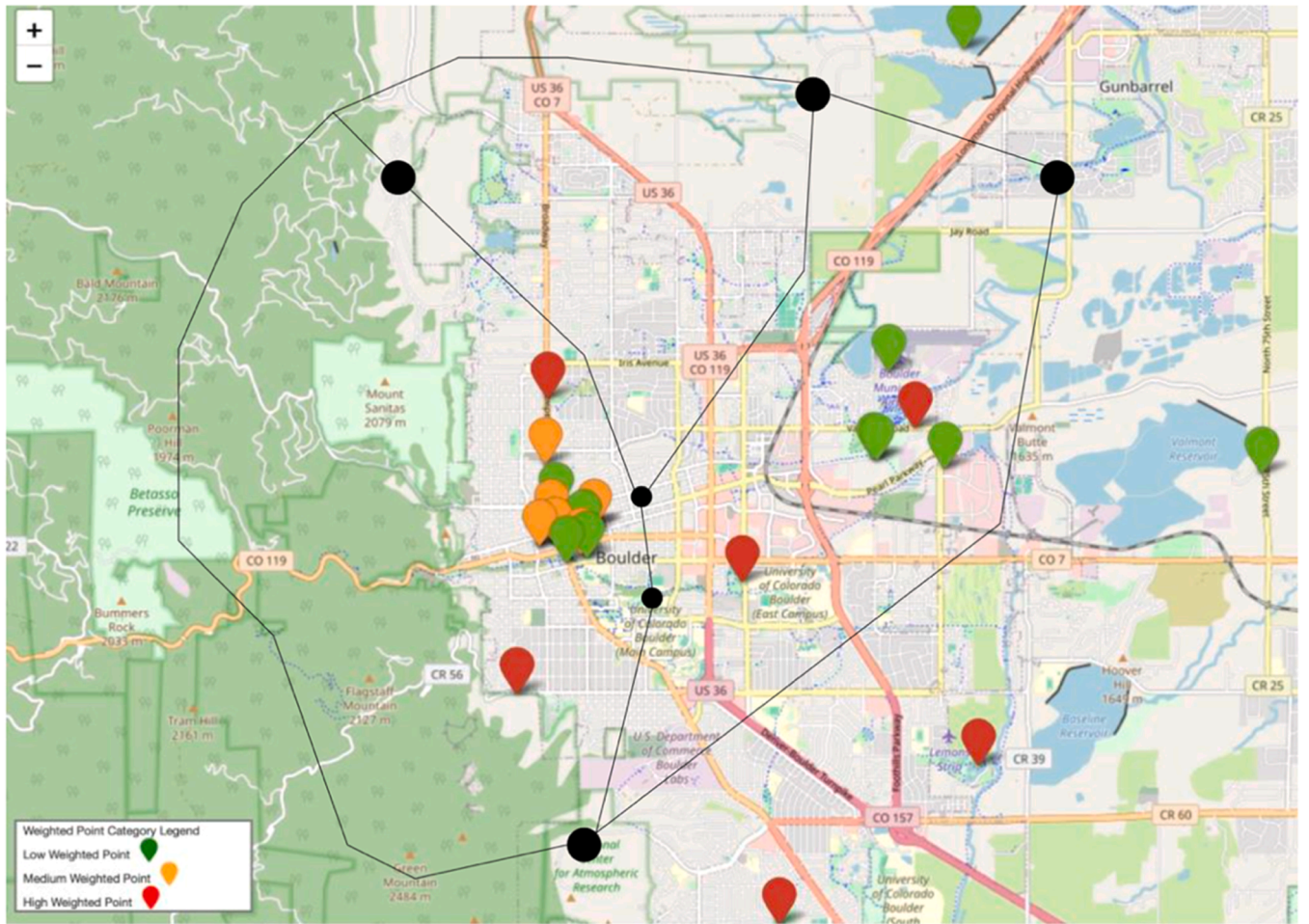


Fig. 14. Power Distribution Network.

ideal value of k .

3.3. K-means clustering algorithm formular

As stated by (Bui et al., 2023), the data for each point u is provided in (1), and the K-means clustering algorithm computes the Euclidean distance to partition the observations into distinct clusters. The Euclidean distance between two data points (x , u) in space is defined by the formula in (2), and the ideal solution is achieved by minimising the total Euclidean distance between the cluster centroid and all data points allocated to that cluster. The optimisation is also demonstrated in the formula presented in (2). The positions of the cluster centroids are determined using the formula specified in (3).

Step 1: Select k at random, assuming

$$u = u_1, u_2, u_3, \dots, u_k \quad (1)$$

where u is the set of all centroid.

Step 2: compute the Euclidean distance towards the centre

$$\min_{ui} \sum_{i=1}^m d(x_i - u_i) \quad (2)$$

where d , x_i , and u_i Euclidean distance, data point and closest point to the centroid

Step 3: Determine the new centroid.

$$u_i = \frac{1}{|s_i|} \sum_{x_i \in s_i} x_i \quad (3)$$

where s_i is set of all point assign to the cluster

Step 4: Repeat step 2 and 3 until it converges

Fig. 3 illustrates that a total of 18 iterations has been employed to elucidate the operational sequence of the k-means clustering method in determining the ideal centroid placements. The k-means clustering process commences with the initialisation of centroids, which are randomly chosen data points from the dataset. During the initial iteration, each data point is allocated to the nearest centroid according to a distance measure, commonly Euclidean distance. This forms clusters, with each data point assigned to the cluster indicated by the nearest centroid. Upon the assignment of all points, the algorithm recalibrates the centroids by calculating the mean of all data points within each cluster.

In successive iterations, the algorithm reiterates the assignment and update phases. Data points are allocated to the nearest centroid, and new centroids are computed based on the revised clusters. This procedure persists until the centroids stabilise, which in this instance occurs after 17 iterations, indicating that their positions no longer exhibit substantial changes or that the assignments of points to clusters remain constant. Upon achieving convergence, the algorithm terminates, producing separate clusters together with their corresponding centroids, prepared for analysis or subsequent processing.

3.4. Custom K-means cluster algorithm for optimization

The algorithm's goal function consists of two elements: the squared Euclidean distance between proposed charging station locations and a penalty term that deters the establishment of new stations near current

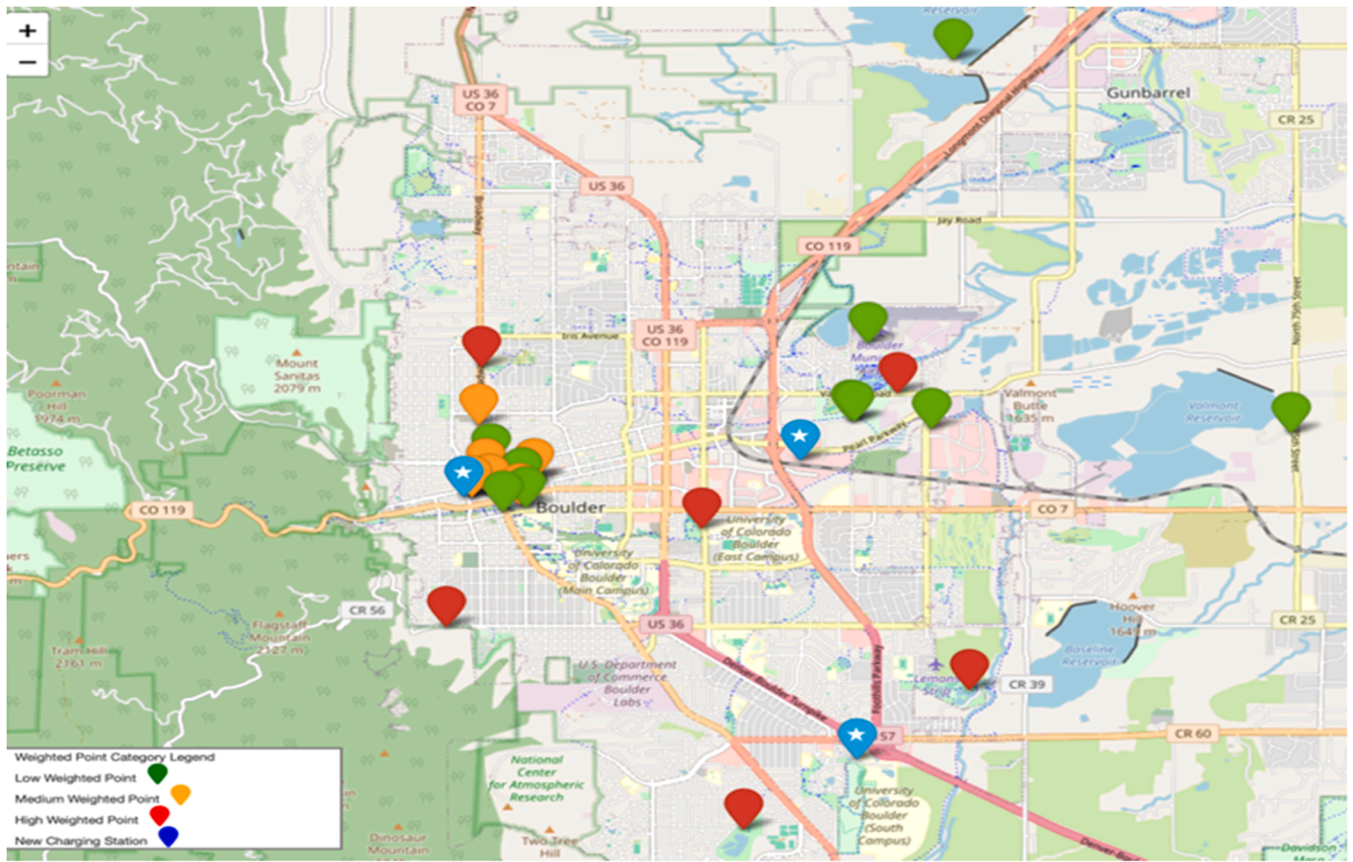


Fig. 15. Using the k-means to optimize the existing station.

low-weighted places. By optimising this goal function, the algorithm determines new charging sites that efficiently support the expected increase in EV use while reducing duplication in the placement of charging stations.

The customised K-Means clustering algorithm enhances conventional K-Means by integrating weights for data points and a penalty term to augment clustering quality. The weighted points (energy demand) guarantee that more significant data points exert a greater impact on cluster formation, while the penalty term deters centroid positioning near low-weighted points to improve the robustness of the clusters (Liu et al., 2023). This method systematically allocates points to clusters and adjusts centroids, reconciling the significance of data points with the necessity to circumvent low-weighted regions. A penalty term was incorporated into the custom k-means clustering algorithm to prevent the relocation of stations near low-weighted charging points for optimal placement. As illustrated in Fig. 4

$$d(\mathbf{x}_i, \mathbf{c}_j) = w_i |\mathbf{x}_i - \mathbf{c}_j|^2 \quad (4)$$

$$P(\mathbf{c}_j) = \alpha \sum_{i=1}^N \frac{1}{w_i + \epsilon} |\mathbf{x}_i - \mathbf{c}_j|^2 \quad (5)$$

where (ϵ) is a small constant to avoid division by zero.

$$J = \sum_{j=1}^K \sum_{i \in S_j} w_i |\mathbf{x}_i - \mathbf{c}_j|^2 + \sum_{j=1}^K P(\mathbf{c}_j) \quad (6)$$

where (S_j) is the set of indices of points assigned to cluster (j) .

Algorithm:

Step 1: Initialize (K) centroids $(\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_K)$.

Step 2: Assign each point $(\mathbf{x}_i)$ to the nearest centroid $(\mathbf{c}_j)$ based on the weighted distance:

$$j = \underset{k}{\operatorname{argmin}} \left(w_i |\mathbf{x}_i - \mathbf{c}_k|^2 + \frac{\alpha}{w_i + \epsilon} |\mathbf{x}_i - \mathbf{c}_k|^2 \right) \quad (7)$$

Step 3: Update each centroid $(\mathbf{c}_j)$ to minimize the objective function (J) :

$$\mathbf{c}_j = \frac{\sum_{i \in S_j} w_i \mathbf{x}_i}{\sum_{i \in S_j} w_i} \quad (8)$$

Step 4: Repeat Steps 2 and 3 until convergence (i.e., centroids no longer change significantly).

Step 5: A penalty term is introduced during this process. It discourages placing centroids (new charging stations) near locations with low demand or existing stations with low usage. The penalty helps to distribute the stations more effectively, ensuring they are placed where they will be most utilized.

$\mathbf{x}_i$ represents the i -th data point, $\mathbf{c}_j$ represents the centroid of the j -th cluster, w_i is the weight assigned to data point $\mathbf{x}_i$ and $|\mathbf{x}_i - \mathbf{c}_j|^2$ denotes the squared Euclidean distance between data point $\mathbf{x}_i$ and centroid $\mathbf{c}_j$. α is a penalty coefficient that controls the influence of the penalty term, N is the total number of data points, ϵ is a small positive constant to avoid division by zero, $\frac{1}{w_i + \epsilon}$ represents the inverse weight of data point $\mathbf{x}_i$, adjusted to ensure stability. The K is the number of clusters and S_j is the set of indices of data points assigned to cluster J . argmin_k denotes the argument of the minimum, i.e., the index j that minimizes the given expression, $w_i |\mathbf{x}_i - \mathbf{c}_k|^2$ is the weighted distance from data point $\mathbf{x}_i$ to centroid $\mathbf{c}_k$, $\frac{\alpha}{w_i + \epsilon} |\mathbf{x}_i - \mathbf{c}_k|^2$ is the penalty term for placing the centroid near the data point $\mathbf{x}_i$. The $\mathbf{c}_j$ is the updated centroid of cluster j , when $\sum_{i \in S_j} w_i \mathbf{x}_i$ is the weighted sum of the data points assigned to cluster j and $\sum_{i \in S_j} w_i$ is the sum of weights of the data points

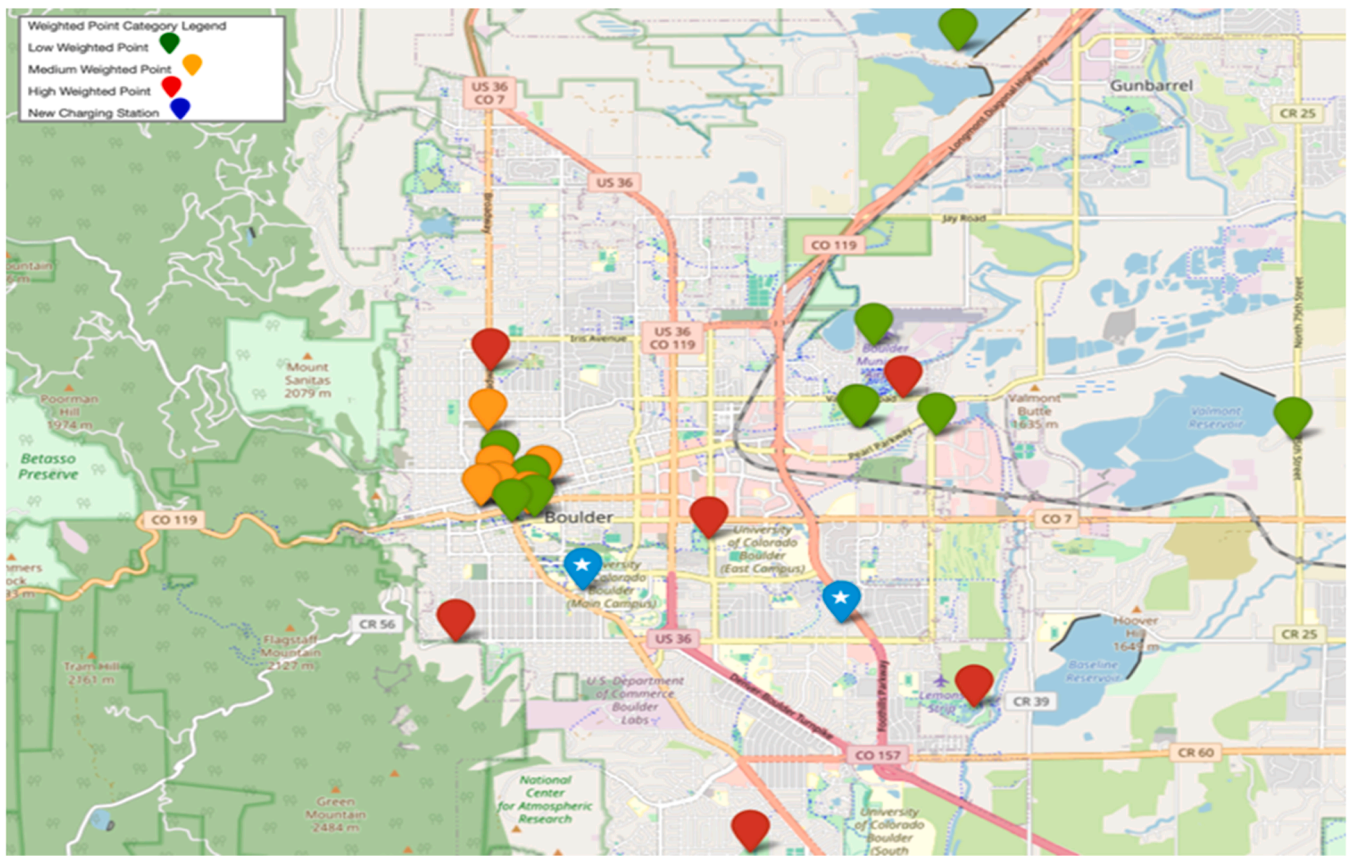


Fig. 16. Optimal locations for existing charging stations custom k means.

assigned to cluster j .

The weighted distance measure adjusts the standard Euclidean distance by incorporating the weight w_i of each data point. This ensures that points with higher importance (higher weights) have a greater influence on the distance calculation, thus affecting the clustering process more significantly (4). The penalty term discourages the placement of centroids near low-weighted points. The inverse weight $\frac{1}{w_i + \epsilon}$ means that points with lower weights contribute more to the penalty, thus pushing centroids away from such points (5).

The objective function combines the weighted distances within clusters and the penalty terms. The goal is to minimize this function, thereby achieving a balance between clustering data points close to their respective centroids (considering their weights) and avoiding centroids near low-weighted points (6).

Each data point is assigned to the cluster whose centroid minimizes the combination of the weighted distance (Liu et al., 2023) and the penalty term. This ensures that the assignment considers both the importance of the data point and the penalty for low-weight locations (7). The centroid of each cluster is updated to be the weighted mean of the data points assigned to that cluster. This update rule ensures that the centroid reflects the distribution of the data points, considering their weights, which leads to more meaningful and representative cluster centers.

4. Results and discussion

4.1. Location and capacity data analysis

Analyzing the data to identify patterns or correlations between the station characteristics (e.g., location, type of station, number of charging points) and the observed energy consumption levels could provide valuable insights into the factors that influence the energy usage at these

charging stations. This could include considerations such as the demographics of the users, the availability of alternative transportation options, or the prevalence of EVs in the surrounding area. Comparing the results of the 2023 with the previous years we tend to see more encouragement of EV users in adoptions recently also the availability of charging stations tends to reduce the traffic at the charging station and also satisfying the charging demand.

Station energy use per kWh in 2018 is shown in Fig. 5. These findings optimise EV charging station placement and capacity for sustainable transportation station. Statistics show that charging stations utilise different amounts of energy, thus more must be put to meet demand. Energy-intensive include Boulder/Baseline St1, Comm Vitality/1000walnut1, 11000walnut1, and 1104 Spruce1. EV use is likely strong here. S1, S2, and Comm Vitality/2200 Broadway1 use less energy. Boulder/Annex St1 has two stations without energy usage data, raising maintenance concerns. Stations close for maintenance or technical reasons, thus observation uses no energy. Depending on usage, Boulder/Annex St1 may be a backup during peak demand or extraordinary occurrences. Recent station installation or remodelling may limit data availability throughout the stated timeframe.

The new Boulder/East Rec and Boulder Junction/Junction St1 stations' 2019 Energy Consumption per Kwh is shown in Fig. 6. Energy use doubled in 2019. Boulder has more EVs and traffic. Boulder/N Since Boulder Rec 1 uses 13,000 kWh annually, it needs an EVCS. Energy-intensive stations include Comm Vitality/1104 Spruce1, 1400 Walnut1, 1000 Walnut1, and 1100 Walnut1. Here, electric car use is likely high. Boulder/ Boulder Park S1 uses less than Comvita/ 2200 Broadway 1. Poor station or electric car use may be attributable. Two Boulder/Annex St1 stations used incredibly little energy this year, raising management concerns. Multiple factors may produce this. Technical issues may have closed the station and limited energy monitoring. Peak demand or unusual events may require Boulder/Annex St1 as a backup.

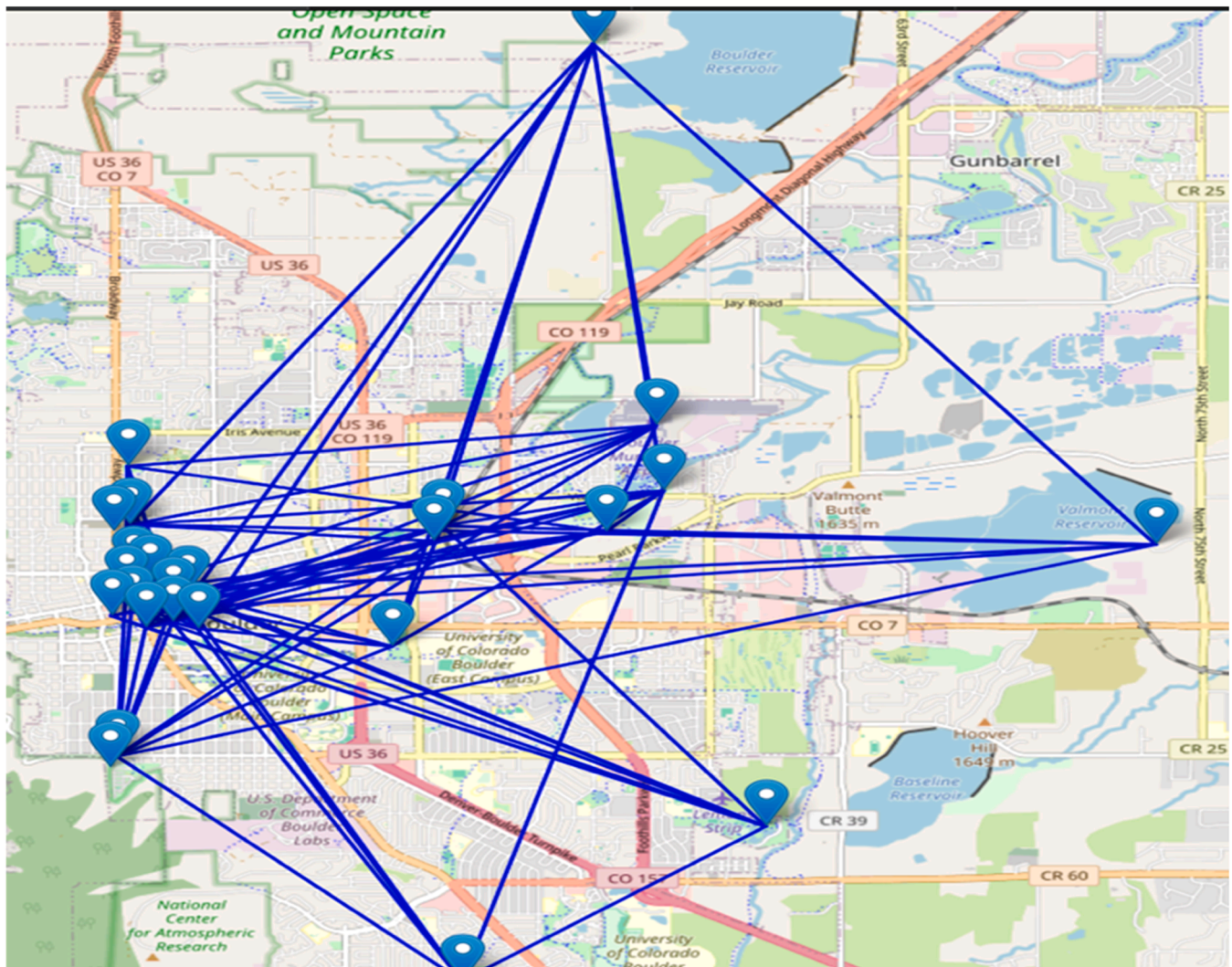


Fig. 17. Road map of the existing charging station in boulder city.

As of the year 2020 a total of 26 charging stations was established in boulder city Fig. 7, compare to the previous years which are 20 stations in 2018 in Fig. 5 and 22 stations in 2019 in Fig. 6. The rate at which the EV adoption rate is increasing is beyond expectations, thus charging demand also increases. Some stations like Boulder/Airport St1 station, Boulder/ Carpenter Park1 Station, Boulder / Reservoir St1 station and some others are established in the year 2020.

Fig. 8 and Fig. 9 illustrate the most widely used EV charging stations need increased capacity or optimization to meet rising demand. Energy consumption patterns vary among stations, influenced by factors like location, accessibility, and local population needs. High energy-consuming sites like Boulder/Carpenter Park2 and Boulder/East Rec require further examination. Moderate consumption stations like Comm Vitality/1104 Spruce and Comm Vitality/1500 Pearl provide benchmarking insights. Low-energy-consuming stations like Boulder/Rec Centre St2 and Boulder/Facilities St1 may reveal potential underutilization opportunities.

Furthermore, the purpose of Figs. 5–10 of this analysis is show a clearer background study, charging demand pattern, EV adoption rate in recent years and the need to optimize the existing charging station for cost minimization, efficient charging distribution and to identify any regional or station-type-specific differences in energy consumption patterns. Such insights could inform policy decisions, grid planning, and the strategic deployment of EV charging station to meet the growing demand for electric mobility while ensuring the reliable and efficient

operation of the power grid.

Fig. 11 shows the cumulative kWh charging demand for electric cars (EVs) from 2018 to 2023. Each bar shows EV charging energy used in a given year, indicating a rising demand. The overall energy was low in 2018, but it rose significantly in 2019, followed by swings in 2020 and 2021. 2023 saw peak charging demand above 350,000 kWh, indicating significant growth in 2022 and 2023. Fig. 11 estimates all Boulder City charging station energy consumption from 2018 to 2023. The picture shows annual city billing demand in each sector. It provides valuable information into EV charging station annual energy usage. The Figure provides a full overview and simplifies annual energy consumption comparisons throughout the specified timeframe.

This Fig. 11 indicates increased EV adoption and EV charging station dependence. EV technology, charging station availability, and environmental consciousness may be driving the ongoing upward trend in electric mobility adoption. The data shows the need to plan for better charging station and energy supply to meet future demand. Results may improve EV charging station optimisation and sustainability. By identifying high-energy usage areas, particular improvements and optimisation can reduce energy waste and boost efficiency. This will make the charging station greener and reduce greenhouse gas emissions.

Fig. 12 depicts the growth in the number of unique EV charging stations from 2018 to 2023. Each bar represents the total count of charging stations available in each year. The figure shows a steady increase over time, starting with a modest number in 2018, followed by a

Table 1
Methodologies in the literature.

Methodology	Ref	Factors
Symbiotic organism Search algorithm	(Chakraborty et al., 2022)	① ②
Dynamic real-time data, Optimal location planning method	(Kong et al., 2019)	⑦(⑧)②
Land use approach, hexagon based, multi-criteria method, Greedy algorithm	(Csiszár et al., 2019)	② ③⑤⑨⑩
Multi-objective approach, hierarchical clustering	(Bitencourt et al., 2021)	③ ①⑥③(⑪)
Genetic algorithm	(Zhou et al., 2022)	②⑥⑧(⑪)(⑬)
Bat algorithm	(Bitencourt et al., 2021)	②①⑥③(⑪)
Queuing method, mixed integer, mathematical method	(Uslu and Kaya, 2021)	②③(⑫)⑥
Meso-level planning	(Torkey and Abdelgawad, 2022)	②③⑦④⑥①⑤
Voronoi diagram	(Qiao et al., 2018)	②⑥⑦③⑨(⑭)
Sioux fall network	(He et al., 2015)	②(⑭)(⑫)(⑬)(⑦)(③)(⑭)
HOMER pro	(Oladigbolu et al., 2022)	②⑥(⑭)③
Chicken swarm, monte Carlo simulation	(Savari et al., 2023)	②(⑭)(⑫)⑥
Nash equilibrium, bi-electric problem	(Savari et al., 2023)	②(⑭)(⑫)⑥
Voronoi, fuzzy-AHP method	(Zheci Tang et al., 2013)	②④⑨⑥(⑭)(⑨)③
Stochastic simulation	(Gang, 2016)	②(⑭)③⑦(⑫)(⑦)
Python, PYPSA, optimal linear	(Meriem Belaid, Hatim, 2022)	④ ③⑥⑤(⑫)(⑪)
Special queuing, stochastic model, discreet model network	(I. Safak Bayram, S. B. 2020)	⑤ ③⑧(⑭)

NOTE: ① EV Adoption rate, EV type, charging technology ② Environmental factors e.g., CO2 emission, climate change, global warming, greenhouse gas emission, noise emission, pollutions, Ambient temperature ③ Charging demand, charging behavior, charging time, queuing time ④ Traffic flow or Traffic pattern, traffic density ⑤ Population Density, high density area ⑥ Renewable energy integration, energy source, Energy consumption, power loss minimization, power consumption, energy efficiency ⑦ Driving behavior, Driver's satisfaction & unsatisfaction, ⑧Cost e.g., construction cost, cost budget and financial limitation, user cost, cost-effectiveness, cost optimization, electric cost ⑨ Geographic constraints Land use, land availability ⑩ Park and ride facilities, ⑪ User convenience, user preference, user perspective (⑫ Distance to charging station (driving range), connectivity of the road network, charging station availability (⑬ Number of charging stations (⑭ Demand area, existing station (⑮ Commercial property index (⑯ Advancement in battery technology, battery residual capacity (⑰ Travel behavior, decision making (⑰ Charging station Economy (operator, incentives & policies, comprehensive benefit, economic availability (⑱ Expert consultation (⑳ Grid load management (peak hour charging), impact on electric power.

Table 2
Assumed location.

Station	Latitude	Longitude	Energy Consumption (weight)
A	40.01	−105.25	1200
B	39.98	−105.27	800
C	40.05	−105.23	1500

Table 3
For Centroid 1 (40.00, −105.24).

Station	Distance without Penalty	Penalty	Total Distance (with Penalty)
A	$(\sqrt{(40.01 - 40.00)^2 + (-105.25 + 105.24)^2} = 0.0141)$	0 (high demand)	0.0141
B	$(\sqrt{(39.98 - 40.00)^2 + (-105.27 + 105.24)^2} = 0.0360)$	$(0.2 \cdot \frac{1}{800 + 0.01} \cdot 0.0360^2 = 0.000002)$	0.036002
C	$(\sqrt{(40.05 - 40.00)^2 + (-105.23 + 105.24)^2} = 0.0500)$	0 (high demand)	0.0500

gradual rise in 2019 and 2020. Notably, the growth accelerates in 2021, continuing to increase significantly through 2022 and reaching its peak in 2023, indicating a clear upward trend in charging station.

This upward trend in the number of charging stations correlates with the increasing demand for EVs, as highlighted in the previous Fig. 11 showing total charging demand. The expansion of charging station is crucial for supporting the growing EV market, as it enhances accessibility and convenience for users. The continuous growth in unique charging stations suggests ongoing investments and efforts to improve the EV ecosystem, making electric mobility more viable and encouraging further adoption among consumers.

4.2. Model implementation

4.2.1. Scenario

Assume we are optimising the location of EV (EV) charging stations in a small city. We have three data points indicating current stations, and the objective is to determine best new charging locations by analysing energy use, charging demand, and proximity to existing stations with a custom K-means clustering method.

4.2.2. Data

We have three operational charging stations with the subsequent coordinates (latitude, longitude) and weights denoting energy consumption (elevated weights signify greater consumption, which should incentivise the establishment of additional stations in proximity):

Our objective is to identify two additional station locations while minimising the distance to high-demand stations and circumventing low-demand areas.

1. Initialization: Randomly select two initial centroids (clusters). Let's pick these:

- Centroid 1: (40.00, −105.24)
- Centroid 2: (40.04, −105.26)

2. Calculate Distance: Using the Euclidean distance formula:

$$d(x,y) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

We'll calculate the distance of each station from the centroids. We also add a penalty term to the stations with lower energy consumption (station B in this case) to discourage positioning near low-demand areas. The penalty term can be calculated using:

$$P(c_j) = \alpha \cdot \frac{1}{w_i + \epsilon} \cdot |x_i - c_j|^2$$

Where (α) is a penalty coefficient, (w_i) is the weight (energy consumption), and (ϵ) is a small constant to avoid division by zero.

Assume:

– $\alpha = 0.2$

– $\epsilon = 0.01$

We will compute the distance for each station and apply the penalty term.

Table 4

For Centroid 2 (40.04, −105.26).

Station	Distance without Penalty	Penalty	Total Distance (with Penalty)
A	$(\sqrt{(40.01 - 40.04)^2 + (-105.25 + 105.26)^2} = 0.0316)$	0 (high demand)	0.0316
B	$(\sqrt{(39.98 - 40.04)^2 + (-105.27 + 105.26)^2} = 0.0620)$	$(0.2 \frac{1}{800 + 0.01} - 0.0620^2 = 0.000006)$	0.062006
C	$(\sqrt{(40.05 - 40.04)^2 + (-105.23 + 105.26)^2} = 0.0316)$	0 (high demand)	0.0316

Table 5

Repositioned location using general k means.

Stations	Latitude	Longitude
Comm Vitality / 1000walnut1	40.02119191	105.24925003
Boulder / Annex St1	40.02162423	105.27960816
MunicipalSc / 2200 Broadway1	39.97479514	105.249062

Table 6

Repositioned location using custom k means.

Stations	Latitude	Longitude
Comm Vitality / 2200 Broadway2	40.08008627	105.2364968
Boulder / Osmf Fleet 2	39.99257048	105.21947216

3. Assignment Step: Now, we assign each station to the nearest centroid (considering the penalty). Based on the distances:

- Station A is closer to Centroid 1.
- Station B is closer to Centroid 1 (despite the penalty).
- Station C is closer to Centroid 2.

4. Update Centroids: Recalculate the centroids by averaging the positions of the stations in each cluster.

- New Centroid 1 (average of A and B):

$$\left(\frac{40.01 + 39.98}{2}, \frac{-105.25 + (-105.27)}{2} \right) = (39.995, -105.26)$$

- New Centroid 2 (only C):

(40.05, −105.23)

5. Iteration: Repeat steps 2–4 with the updated centroids until convergence (the centroids no longer move significantly).

Final Result: After a few iterations, the centroids converge, and the algorithm suggests placing the new stations near:

- Centroid 1: (39.995, −105.26)
- Centroid 2: (40.05, −105.23)

These locations represent the optimal placement of new EV charging stations considering both energy consumption and the penalty for low-demand areas.

Furthermore, the insights derived from this analysis may yield practical applications for the construction and development of charging stations in Boulder City and analogous areas. Strategically positioning charging stations in areas of optimised energy usage enhances the convenience and accessibility for EV drivers.

Boulder City's diverse EV charging stations are shown in Fig. 13. The map's points represent charging stations. EV owners can charge at each site. We can determine charging station placement by analysing point distribution. Certain localities may have more charging stations because

some points are pooled. This may be due to high population density, EV use, or strategic placing near notable locations. However, more dispersed locations may indicate a more uniform citywide charging station distribution. These stations may be strategically placed to give Boulder City EV owners easy access. Dots on the Fig. 13 cover Boulder City's EV charging stations in various locations. Electric car owners need this information to plan journeys and find charging outlets.

Energy-intensive charging facilities are marked red on the Fig. 13. These stations use more energy than others. Boulder/East Rec, Boulder/Carpenter Park2, and Boulder/Park1 are energy-intensive stations. Understanding the causes of high energy usage at these sites is crucial to improving charging network efficiency. The Fig. 13 shows medium-energy charging stations as orange markers. Moderate energy use at these stations may reveal network usage trends. Comm Vitality/1104 Spruce1, /1500 Pearl1, and /2200 Broadway1 are medium-energy-consuming stations.

Finally, green markers in Fig. 13 indicators indicate low-energy charging stations. These stations consume little energy, suggesting underutilisation or optimisation potential. Boulder/Rec Centre St2 and Boulder/Facilities St1 are low-energy stations. The distribution of charging stations can also help authorities identify areas where more stations are needed to increase EV accessibility and convenience.

Fig. 14 shows The location information for substations in Boulder city of Colorado, was obtained from Open Station Map and overlaid onto an existing transportation map. The substations in this area are mainly categorized into two types: black dots represent the geographic locations of the substations, with larger dots indicating substations of 115 kV and above, and smaller dots representing substations with a capacity of 33 MW. Black lines show the connections between the substations.

Fig. 15 shows The K-means clustering-relocated charging stations are marked in blue. The K-means algorithm was used to find Boulder City's optimum charging station relocation locations. This will improve charging station efficiency and strategic positioning to match regional EV customers' energy demand and usage patterns. The blue markers represent the relocated locations likely suggest regions with similar energy consumption trends where charging stations can be optimally constructed to fulfil local EV user needs.

The Fig. 15 colour-coded markers show charging station energy consumption. The red marks indicate stations with high energy usage, which have the largest charging demand. Stations with medium and low energy use are orange and green, indicating a more balanced or reduced usage pattern. By moving stations to blue-marked regions, the system can better meet local electric car users' energy consumption patterns. This may reduce energy waste, improve charging station accessibility, and boost charging network efficiency.

Blue markers show new charging station locations in Fig. 16. A custom k-means clustering algorithm was used to identify the best relocation of Boulder City's charging stations to address coverage gaps and ensure quick access for EV users. The custom k-means clustering method is needed to find the best station relocation sites. This algorithm considers charging demand, electric car usage, and geographic dispersion. The program strategically selects station sites to satisfy EV owners' needs by analysing these parameters.

The algorithm evaluates multiple locations, including those with reduced energy usage, by using energy consumption as a weighted factor. These regions may have low EV users, reducing charging demand

and achieving an environmentally friendly charging network. Energy use analysis helps optimise charging station placements, improving electric car performance and sustainability. These locations were carefully selected to optimise charging station distribution across the city, ensuring EV customers had convenient access. The algorithm reduces charging station distance from high-demand EV zones.

A penalty term prevents the algorithm from shifting charging stations near low-weighted points. This prevents regions with low charging demand or low energy efficiency from being prioritised. The algorithm avoids some sites to promote more equal charging station placement, improving network sustainability and efficiency. The map shows new charging station locations and emphasises the relevance of k-means clustering in improving them. The algorithm's iterative computations and cluster allocations reduce intra-cluster distances and increase inter-cluster distances. This ensures charging station distribution, reducing EV owner travel distances and promoting citywide fairness.

The algorithm's optimal charging station location, shown by blue marks, shows its ability to satisfy high EV charging demand and provide Boulder City EV owners with a complete, efficient, and equitable network. The custom K-means clustering technique efficiently finds new station sites while avoiding redundancy, emphasising the need of data-driven station planning. Adding a penalty term to the optimisation process ensures station placement that considers proximity to existing stations. These methods address current demand and predict electric car growth, which is essential for sustainable urban development.

The blue lines on the map denote the pathways that EVs may utilise to reach the specified charging stations. This illustration offers significant insights regarding the connectivity and accessibility of charging stations throughout the city. Fig. 17 illustrates the current stations and the prospective paths available for EV users to access charging stations efficiently. Comprehending the road network is essential for evaluating the efficacy and efficiency of the charging station network (He et al., 2015). To evaluate the distribution and accessibility of charging stations around Boulder City. It offers a visual depiction of the distribution of charging stations across the city, emphasising regions with potentially restricted access to these facilities. This connection analysis identifies potential deficiencies or locations where additional charging stations may be necessary to offer extensive coverage and continuous access for EV users.

Analysing the road network, researchers may evaluate traffic flow and congestion around the charging stations. This information is essential for optimising the placement of charging stations and detecting places where possible traffic problems may emerge owing to increased EV usage. The road network visualisation facilitates the identification of ideal routes for EV owners to access charging stations.

Fig. 17 establishes a basis for the future development of charging stations in Boulder City. By pinpointing regions with inadequate coverage or elevated demand, stakeholders may make informed decisions about the installation of additional charging stations and potential enhancements to the current road station. This study's results illustrate the efficacy of the bespoke K-means clustering algorithm in enhancing the placement and capacity of EV charging stations.

4.3. Utilization-based optimization and optimal charging station locations

Conducted a detailed analysis of station utilization patterns to identify underutilized or inefficiently used charging stations. This can inform decisions on relocating, repurposing, or decommissioning certain stations to reallocate resources to areas with higher demand. Regularly monitor and analyze usage data to identify opportunities for optimizing the charging network, such as adjusting station capacities, modifying pricing structures, or implementing targeted marketing and awareness campaigns to drive increased utilization

The custom K-means clustering algorithm identified several optimal locations for new charging stations in Boulder City, as shown in Fig. 16 and Table 5. These locations were strategically repositioned based on

factors such as charging demand, energy consumption and areas with high concentrations of EV owners.

The distribution of the recommended locations throughout the city help to ensure convenient access to charging facilities for EV owners, regardless of their location within Boulder City. Incorporating energy consumption as a weighted factor, the algorithm considered areas with lower energy consumption, potentially indicating lower EV usage and charging demand, aligning with the goal of creating an environmentally friendly and efficient charging network.

4.4. Road network analysis

Fig. 17 provides an assessment of the coverage and accessibility of charging stations across Boulder City, considering the road network. This analysis aids in identifying potential gaps or areas where additional charging stations may be required to ensure comprehensive coverage and uninterrupted access for EV owners. Analyzing the road network, researchers can assess traffic flow and congestion around charging stations, optimize placement to minimize travel time for EV owners, and identify potential traffic issues arising from increased EV usage. This information is crucial for future expansion planning and potential upgrades to the existing road network to accommodate the growing demand for EV charging services.

The results of this study emphasize the importance of strategically placing charging stations to ensure easy and convenient access for EV owners, ultimately contributing to increased adoption of EVs and reducing greenhouse gas emissions. However, it is essential to acknowledge the limitations of the study, such as uncertainties in data analysis and modeling techniques, and the need for real-world case studies to further enhance the effectiveness of EV charging station optimization. Considering factors such as traffic patterns, distance, and road conditions, researchers can provide recommendations to improve the efficiency of the charging stations and minimize travel time for EV owners.

5. Conclusion

This work presents a tailored k-means clustering technique aimed at optimising the placement of existing EV charging stations. Multiple criteria, including EV adoption, charging demand, and energy usage, were examined to assist researchers in identifying optimal sites for charging stations and determining the suitable placement for each area. The incorporation of the penalty term into the custom K-means clustering algorithm has demonstrated significant value in the optimisation process. This research underscores the necessity of strategically locating charging stations to facilitate easy and convenient access for EV drivers. Enhancing the placement of charging stations based on charging demand will augment their convenience and efficiency, potentially resulting in greater EV adoption. This can subsequently aid in diminishing greenhouse gas emissions and enhancing air quality. The Custom K-means clustering method proficiently determines ideal locations for new stations while reducing redundancy, underscoring the importance of data-driven decision-making in station planning. Furthermore, the inclusion of a penalty term in the optimisation process guarantees prudent station placement that takes into account proximity to existing stations. This technique not only meets present demand but also forecasts future EV growth, which is vital for sustainable urban development.

However, it is imperative to recognise the constraints of this study. The real-world application of the suggested optimisation solutions may face practical obstacles that were not addressed in the research. Subsequent research may expand upon these findings by incorporating supplementary parameters and executing empirical case studies across diverse geographical contexts to further improve the efficacy of EV charging station optimisation.

Author statement

We declare that the work titled "Location Optimization of EV Charging Stations: A Custom K-Means Cluster Algorithm Approach" is original, has not been previously published, and is not under consideration for publication elsewhere. All authors have read and approved the final manuscript and have made significant contributions to the research, including its conceptualization, data collection, analysis, and manuscript preparation.

In this study, we address Electric Vehicle Charging Station Location Optimization using a Customized K Means clustering algorithm. The research outcomes contribute to the advancement of knowledge in the field of Electric Vehicle Charging, as the locations of the existing charging stations in the study area are optimized using the energy demand factor.

There are no conflicts of interest, and all authors comply with Energy Reports' ethical guidelines. Proper acknowledgements have been made, and no funding has been received.

We appreciate your consideration of our manuscript for proceeding with the review stage in Energy Reports.

CRediT authorship contribution statement

Muhammad Rabiul Abdullahi: Writing – original draft, Visualization, Validation, Software, Formal analysis. **Qing-Chang Lu:** Validation, Supervision, Project administration, Conceptualization. **Sajib Tripura:** Validation, Resources, Investigation, Data curation. **Adil Hussain:** Writing – review & editing, Validation, Methodology, Investigation, Conceptualization. **Peng-Cheng Xu:** Validation, Investigation, Formal analysis. **ShiXin Wang:** Investigation, Validation, Visualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

Thanks to the writing workshop of the International Education School of Chang'an University.

Data availability

Data will be made available on request.

References

- Afshar, S., Macedo, P., Mohamed, F., Disfani, V., 2021. Mobile charging stations for EVs – a review. *Renew. Sustain. Energy Rev.* 152. <https://doi.org/10.1016/j.rser.2021.111654>.
- Bitencourt, L., Abud, T.P., Dias, B.H., Borba, B.S.M.C., Maciel, R.S., Quirós-Tortós, J., 2021. Optimal location of EV charging stations in a neighborhood considering a multi-objective approach. *Electr. Power Syst. Res.* 199. <https://doi.org/10.1016/j.epsr.2021.107391>.
- Bouguerra, S., Bhar Layeb, S., 2019. Determining optimal deployment of EVs charging stations: case of Tunis City, Tunisia. *Case Stud. Transp. Policy* 7 (3), 628–642. <https://doi.org/10.1016/j.cstp.2019.06.003>.
- Bui, V.-H., Hussain, A., Zarrabian, S., Kump, P.M., Su, W., 2023. Clustering-based optimal operation of charging stations under high penetration of EVs. *Sustain. Energy, Grids Netw.* 36. <https://doi.org/10.1016/j.segan.2023.101178>.
- Carra, M., Maternini, G., Barabino, B., 2022. On sustainable positioning of EV charging stations in cities: an integrated approach for the selection of indicators. *Sustain. Cities Soc.* 85. <https://doi.org/10.1016/j.scs.2022.104067>.
- Chakraborty, R., Das, D., & Das, P. (2022). *Optimal Placement of EV Charging Station with V2G Provision using Symbiotic Organisms Search Algorithm* 2022 IEEE International Students' Conference on Electrical, Electronics and Computer Science (SCEECs),
- Chen, R., Qian, X., Miao, L., Ukkusuri, S.V., 2020. Optimal charging facility location and capacity for EVs considering route choice and charging time equilibrium. *Comput. Oper. Res.* 113. <https://doi.org/10.1016/j.cor.2019.104776>.
- Csiszár, C., Csonka, B., Földes, D., Wirth, E., Lovas, T., 2019. Urban public charging station locating method for EVs based on land use approach. *J. Transp. Geogr.* 74, 173–180. <https://doi.org/10.1016/j.jtrangeo.2018.11.016>.
- Erbas, M., Kabak, M., Özceylan, E., Çetinkaya, C., 2018. Optimal siting of EV charging stations: a GIS-based fuzzy Multi-Criteria Decision Analysis. *Energy* 163, 1017–1031. <https://doi.org/10.1016/j.energy.2018.08.140>.
- Gang, T. (2016). Research on the location and capacity of EV charging stations.pdf.
- Guler, D., Yomralioglu, T., 2020. Suitable location selection for the EV fast charging station with AHP and fuzzy AHP methods using GIS. *Ann. GIS* 26 (2), 169–189. <https://doi.org/10.1080/19475683.2020.1737226>.
- He, F., Yin, Y., Zhou, J., 2015. Deploying public charging stations for EVs on urban road networks. *Transp. Res. Part C: Emerg. Technol.* 60, 227–240. <https://doi.org/10.1016/j.trc.2015.08.018>.
- Kong, W., Luo, Y., Feng, G., Li, K., Peng, H., 2019. Optimal location planning method of fast charging station for EVs considering operators, drivers, vehicles, traffic flow and power grid. *Energy* 186. <https://doi.org/10.1016/j.energy.2019.07.156>.
- Liu, J., Cao, F., Gao, X.-Z., Yu, L., Liang, J., 2023. A cluster-weighted kernel k-means method for multi-view clustering. *Proc. AAAI Conf. Artif. Intell.*
- Liu, H., Li, Y., Zhang, C., Li, J., Li, X., Zhao, Y., 2022. EV charging station location model considering charging choice behavior and range anxiety. *Sustainability* 14 (7). <https://doi.org/10.3390/su14074213>.
- Lu, Q.-C., Xu, P.-C., Zhang, J., 2021. Station-based transportation network vulnerability modeling and analysis. *Phys. A: Stat. Mech. its Appl.* 584. <https://doi.org/10.1016/j.physa.2021.126350>.
- Mastoi, M.S., Zhuang, S., Munir, H.M., Haris, M., Hassan, M., Usman, M., Bukhari, S.S.H., Ro, J.-S., 2022. An in-depth analysis of EV charging station station, policy implications, and future trends. *Energy Rep.* 8, 11504–11529. <https://doi.org/10.1016/j.egyrs.2022.09.011>.
- Meriem Belaid, S.E.B., Hatim Anas (2022). *Optimization Tool of EVs Charging Station.pdf* IFAC CONFERENCE,
- Mishra, S., Verma, S., Chowdhury, S., Gaur, A., Mohapatra, S., Dwivedi, G., Verma, P., 2021. A comprehensive review on developments in EV charging station station and present scenario of India. *Sustainability* 13 (4). <https://doi.org/10.3390/su13042396>.
- Oladigbolu, J.O., Mujeeb, A., Imam, A.A., Rushdi, A.M., 2022. Design, technical and economic optimization of renewable energy-based EV charging stations in Africa: the case of Nigeria. *Energies* 16 (1). <https://doi.org/10.3390/en16010397>.
- Qiao, Y., Huang, K., Jeub, J., Qian, J., Song, Y., 2018. Deploying EV charging stations considering time cost and existing station. *Energies* 11 (9). <https://doi.org/10.3390/en11092436>.
- Sadeghi Ahangar, S., Abazari, S.R., Rabbani, M., 2022. A region-based model for optimizing charging station location problem of EVs considering disruption - a case study. *J. Clean. Prod.* 336. <https://doi.org/10.1016/j.jclepro.2022.130433>.
- I. Safak Bayram, S. B (2020). *Location Analysis of EV Charging Stations for Maximum Capacity and Coverage.pdf*.
- Savari, G.F., Sathik, M.J., Raman, L.A., El-Shahat, A., Hasanien, H.M., Almkhles, D., Abdel Aleem, S.H.E., Omar, A.I., 2023. Assessment of charging technologies, station and charging station recommendation schemes of EVs: a review. *Ain Shams Eng. J.* 14 (4). <https://doi.org/10.1016/j.asej.2022.101938>.
- Shareef, H., Islam, M.M., Mohamed, A., 2016. A review of the stage-of-the-art charging technologies, placement methodologies, and impacts of EVs. *Renew. Sustain. Energy Rev.* 64, 403–420. <https://doi.org/10.1016/j.rser.2016.06.033>.
- Torkey, A., Abdelgawad, H., 2022. Framework for planning of EV charging station: Where should cities start? *Transp. Policy* 128, 193–208. <https://doi.org/10.1016/j.tranpol.2022.09.015>.
- Torres Franco, S., Durán Tovar, I.C., Suárez Pradilla, M.M., Marulanda Guerra, A., 2021. EV charging stations' location in urban transportation networks: a heuristic methodology. *IET Electr. Syst. Transp.* 11 (2), 134–147. <https://doi.org/10.1049/els2.12011>.
- Uslu, T., Kaya, O., 2021. Location and capacity decisions for electric bus charging stations considering waiting times. *Transp. Res. Part D: Transp. Environ.* 90. <https://doi.org/10.1016/j.trd.2020.102645>.
- Yang, J., Dong, J., Hu, L., 2017. A data-driven optimization-based approach for siting and sizing of electric taxi charging stations. *Transp. Res. Part C: Emerg. Technol.* 77, 462–477. <https://doi.org/10.1016/j.trc.2017.02.014>.
- Yang, M., Zhang, L., Dong, W., 2020. Economic benefit analysis of charging models based on differential ev charging station subsidy policy in China. *Sustain. Cities Soc.* 59. <https://doi.org/10.1016/j.scs.2020.102206>.
- Zhan, W., Wang, Z., Zhang, L., Liu, P., Cui, D., Dorrell, D.G., 2022. A review of siting, sizing, optimal scheduling, and cost-benefit analysis for battery swapping stations. *Energy* 258. <https://doi.org/10.1016/j.energy.2022.124723>.
- Zheci Tang, C.G., Pengxin, Hou, Yubo, Fan, 2013. *Optim. Siting EV Chrg. Stations Based Voronoi Diagr. FAHP Method. pdf.* <https://doi.org/10.4236/epe.2013.54B266>.
- Zhou, G., Zhu, Z., Luo, S., 2022. Location optimization of EV charging stations: based on cost model and genetic algorithm. *Energy* 247. <https://doi.org/10.1016/j.energy.2022.123437>.